

PROFILE SUMMARY

Detail-oriented Mechanical Design Engineer with 7+ years of expertise in static and rotary equipment design, 3D modelling, and R&D across oil & gas and industrial sectors. Expertise in Innovating New concepts, 3D Modelling, Developing pressure vessels, heat exchangers, dosing systems, and custom skid packages. Skilled in CAD tools (NX, AutoCAD, SolidWorks), Design calculations (PVElite, AMETank), and CNC programming. Adept at optimizing material use, reducing manufacturing costs, and leading cross-functional project teams. Strong knowledge of ASME, TEMA, and API standards. Skilled in reverse engineering, ERP systems, and basic SAP. Proven leadership managing a 5-member team to meet critical project timelines. Known for strong analytical and interpersonal skills, and a keen ability to interpret mechanical systems and internal components through visual and auditory cues.

AREA OF EXPERTISE

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| <ul style="list-style-type: none">• Design Engineering• Static & Rotary Equipment Design• Reverse Engineering• 3D Modelling & CAD Drafting• Fabrication and Machining Knowledge• Product Development & Detailing• BOM Generation• Piping Design and Layout• PID and ISO Creation | <ul style="list-style-type: none">• General Arrangement (GA) Drawing• Design Calculations• Material Cost Optimisation• CNC Programming• Pressure Vessels and Tank Design• Project Coordination• R&D and Innovation• Team Leadership• Cross-functional Collaboration | Standards & Codes Knowledge <ul style="list-style-type: none">• ASME B16.5, B31.3, B31.8• ASME Sec VIII Div.1• TEMA (Tubular Exchanger Manufacturers Association)• API 650 (Storage Tanks)• API 675 (Metering Pumps)• API 6D / ISO 14313• ASME B16.47 |
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TECHNICAL SKILLS

• NX10/ NX11 • PV-Elite (Pressure Vessel & Heat Exchanger) • Code CAL • STAAD PRO (BASIC) • Pro/E Wildfire 4 (BASIC) • Solid Works & Simulation • Plant 3D • AutoCAD • Hyper Mesh (BASIC) • AMETank ME 11.7 (API 650 Tanks) • 3d Printing Software • Microsoft Office and Internet Applications

WORK HISTORY

GECO Legend Engineering service Pvt. Ltd., Coimbatore, India

Feb 2023 - Present

Senior Design Engineer (Static and Rotary Equipment)

- Engineered various heat exchangers using PVElite software per ASME and TEMA standards.
- Created GA and fabrication drawings for client approvals and manufacturing.
- Performed thermal, mechanical, and structural calculations ensuring design integrity.
- Collaborated with multidisciplinary teams to review materials and construction strategies.
- Addressed client feedback through iterative design modifications.
- Coordinates with client for End User meeting and discussion
- Delivered detailed designs for pressure vessels and ensured regulatory compliance.
- Designed API 650 crude oil storage tanks using AMETANK, including vents, davit arms, and counterweight systems.
- Drafted and issued fabrication and construction drawings aligned with project specs and local environmental codes.
- 3D modelling of various Parts of Agricultural vehicles and Issue for fabrication.
- 3D modelling and Simulation check using Solidworks for GUN cabinet and GUN Safe Locks
- Coordinated with project and fabrication teams to ensure seamless execution.

Key Projects:

- Heat Exchangers & Pressure Vessels – (M/s , L&T, LR Fab.)
- API 650 Crude Oil Storage Tanks – (Vijay Tanks, Adnoc)
- Gun cabinet and Gun Safe – (liberty Safe)

Senior Design Engineer (R&D)

- Led design and R&D for chemical dosing packages in oil & gas and industrial applications.
- Formulated detailed 3D models, piping layouts, and fabrication drawings per client specs.
- Conducted testing and evaluation of newly developed skids.
- Enhanced system performance through continuous improvement and prototyping.
- Worked across mechanical design and fluid system integration for offshore and onshore projects.

Key Projects:

- Biocide Dosing Skid – BPCL
- RON Booster Dosing Skid – IOCL
- Portable Methanol Dosing Skid – ONGC / Tata Projects Ltd
- DRA & CI Dosing System – HPCL
- Chemical Dosing Packages – Offshore Rig – ONGC
- Chemical Dosing Skid – KLJ Petro Plash

Dencil Pumps & Systems Pvt. Ltd., Mumbai, India

Jun 2017 – Jan 2021

Junior Design Engineer (R&D)

- Developed 3D models of dosing systems, isometric pipelines, and assemblies.
- Produced GA drawings, P&ID diagrams, and BOMs based on customer requirements.
- Assisted production in fabrication, machining, assembly, and testing activities.
- Programmed CNC vertical milling machines (SIEMENS 828D M) using NX CAM.
- Supervised project task allocations and design workflows across teams.
- Translated client requirements into functional 3D models and detailed engineering documentation.
- Applied GD&T principles and conducted design reviews with clients.
- Demonstrated practical expertise in lathe, milling, drilling, and welding operations.
- Delivered tailored piping solutions across various industry sectors including oil & gas.

Key Projects:

- Passivation Portable Dosing System – IOCL, Mathura Refinery
- DRA Injection Skid (Containerized) – BORL
- Portable Zytex Dosing System – Zytex Industries
- DRA Dosing System – HPCL
- DRA & CI Dosing Systems – L&T (Hydrocarbon Project)
- Alkali, Acid, Alum Dosing Systems – OTOKLIN Global Business Ltd
- pH Correction Dosing Pump – BASF Petronas Chemicals (BPC Steam Supply)
- CI Dosing System – ESSAR Oil

R&D CONTRIBUTIONS

- Redesigned pump bodies and internal assemblies to optimize manufacturability, significantly reducing material costs and machining time.
- Revamped internal mechanical components across multiple pump types to enhance efficiency and ease of production
 - Mechanical Operated Diaphragm Pumps
 - Hydraulic Operated Diaphragm Pumps
 - Mechanical Operated Single & Double Acting Plunger Pumps
 - Triplex Pumps
- Engineered an LPG Cylinder Auto Washing System for IOCL bottling plants, improving automation and operational throughput.
- Developed a Gun Safe Lock Mechanism and Gun Cabinet, integrating enhanced safety and locking features for secure storage.

EDUCATION**Master of Technology (M. Tech)** in **Design Engineering** from BITS Pilani (Work Integrated Learning Program), India**2023****Bachelor of Technology (B. Tech)** from Dr.MGR Educational & Research Institute, Chennai, India**2017**